value of r can be obtained by taking the term t_{r+1} as the required term. To find the **term independent of x** (term without x), equate the power of x in t_{r+1} to zero, we get the value of 'r'. By substituting the value of r in t_{r+1}, we get the term independent of x.

NOTE

$(x+a)^n = nC_0x^n a^0 + nC_1x^{n-1}a^1$
$+ nC_2x^{n-2}a^2 + ... + nC_rx^{n-r}a^r + ... + a^n \ldots (1)$

(i) To find $(x-a)^n$, replace by $-a$

$(x-a)^n = nC_0x^n a^0 + nC_1x^{n-1}(-a)$
$+ nC_2x^{n-2}(-a)^2 + ...$
$+ nC_rx^{n-r}(-a)^r + ... + (-a)^n$

$= nC_0x^n a^0 - nC_1x^{n-1}(a) + nC_2x^{n-2}(a)^2 + \ldots$
$+ (-1)^r nC_rx^{n-r}(a)^r + ... + (-1)^n(a)^n \quad ...(2)$

Note that we have the signs alternatively.

(ii) If we put $a = 1$ in (1), we get

$(1+x)^n = 1 + nC_1x + nC_2x^2 + ...$
$+ nC_rx^r + ... + nC_nx^n \quad ...(3)$

(iii) If we replace $xby - x$ in (3), we get

$(1-x)^n = 1 - nC_1x + nC_2x^2 - \ldots$
$+ nC_r(-1)^r x^r + ...$
$+ nC_n(-1)^n x^n$

Example 2.39

Expand $(2x + 3y)^5$ using binomial theorem.

Solution

$(x+a)^n = nC_0x^n a^0 + nC_1x^{n-1}a^1$
$+ nC_2x^{n-2}a^2 + ... + nC_rx^{n-r}a^r$
$+ ... + nC_{n-1}x^1 a^{n-1} + nC_nx^0 a^n$

$(2x+3y)^5 = 5C_0(2x)^5 + 5C_1(2x)^4(3y) +$
$5C_2(2x)^3(3y)^2 + 5C_3(2x)^2(3y)^3$
$+ 5C_4(2x)(3y)^4 + 5C_5(3y)^5$

$= 32x^5 + 5(16)x^4(3y) +$
$\frac{5\times4}{2\times1}(8x^3)(9y^2) +$
$5(2x)81y^4 + 243y^5$

$= 32x^5 + 240x^4y + 720x^3y^2$
$+ 1080x^2y^3 + 810xy^4 + 243y^5$

Example 2.40

Using binomial theorem, expand $\left(x^2 + \frac{1}{x^2}\right)^4$

Solution

$\left(x^2 + \frac{1}{x^2}\right)^4 = (x^2)^4 + 4C_1(x^2)^3\frac{1}{x^2}$
$+ 4C_2(x^2)^2\left(\frac{1}{x^2}\right)^2 + 4C_3(x^2)\left(\frac{1}{x^2}\right)^3$
$+ 4C_4\left(\frac{1}{x^2}\right)^4$

$= x^8 + 4x^4 + 6 + \frac{4}{x^2} + \frac{4}{x^4} + \frac{1}{x^8}$

Example 2.41

Using binomial theorem, evaluate $(101)^5$

Solution

$(101)^5 = (100 + 1)^5$

$= (100)^5 + 5C_1(100)^4 + 5C_2$
$(100)^3 + 5C_3(100)^2 + 5C_4$
$(100) + 5C_5$

$= 10000000000 + 5(100000000)$
$+ 10(1000000) + 10(10000)$
$+ 5(100) + 1$

$= 10000000000 + 500000000 +$
$10000000 + 100000 + 500 + 1$

$= 10510100501$

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